Formalizing AI Scientist Systems: A Unified Framework for Automated Scientific Discovery

Jaesol Shin
WeDataLab
ysys143@wedatalab.com

May 2026

Abstract

The emergence of end-to-end systems capable of automating hypothesis generation, experiment execution, and knowledge synthesis represents a significant change in how science is conducted. Despite rapid empirical progress, *AI Scientist* systems lack a unifying formal framework, which makes principled comparison across architectures and scientific domains difficult. We propose SERVO, a six-component framework (S, G, E, V, M, π) comprising the **S**earch space, hypothesis **G**enerator, **E**xperiment executor, **V**alidator, **M**emory, and search **P**olicy (π), grounded in the theory of Partially Observable Markov Decision Processes (POMDPs) and Bayesian Experimental Design (BED). We apply this framework to analyze core AI Scientist systems and their instantiations across seven scientific domains (mathematics, physics, chemistry, materials science, biology, social science, and computer science), and separately examine a complementary artifact infrastructure proposal that addresses the output format and verification layer of such systems. Our analysis surfaces three persistent open problems: (1) the non-automatability of novelty evaluation, which remains dependent on human judgment despite multi-layer automated proxies; (2) the absence of BED-optimal search policies, despite the theoretical availability of tractable approximations; and (3) the underdevelopment of semantic memory M_s, whose absence prevents accumulated experimental experience from improving future search. We show that the completeness of the validator V is the single most consequential design variable: it determines whether closed-loop autonomous operation is feasible in each domain, and current AI Scientist systems can be read as a progression toward more complete validators.

1 Introduction

AI Scientist systems—agents that independently generate hypotheses, execute experiments, and synthesize knowledge—have proliferated rapidly [Boiko et al., 2023, Lu et al., 2024, 2026, Schmidgall et al., 2025, Feng et al., 2026, Liu et al., 2026], yet the field lacks a shared formal vocabulary. Each system introduces bespoke terminology, making systematic comparison difficult. Three questions that bear directly on deployment remain unanswered:

What determines whether a system can operate in a closed loop? Why does autonomous novelty evaluation fail consistently? What is the theoretically optimal search policy?

We address these gaps with three contributions: (1) SERVO, a formal framework (S, G, E, V, M, π) connecting AI Scientist systems to POMDP and Bayesian Experimental Design theory (Section 4); (2) analysis of four core systems, an artifact infrastructure proposal, and domain-specific instantiations across seven scientific fields (Sections 5, 6); and (3) three open problems that constrain progress across all current systems (Section 8).

Scope. We cover systems whose primary goal is automated scientific *discovery* through hypothesis-experiment-validation cycles, excluding specialized tools within human-conducted pipelines [Jumper et al., 2021].

2 Background

Following Kaelbling et al. [1998], we model AI Scientist systems as *partially observable Markov decision processes*: the true state s (unknown structure of nature) is inaccessible; the system maintains a belief $b(s)$ updated by experimental observations. Formally, an AI Scientist POMDP is the tuple $(\mathcal{S}, \mathcal{A}, T, R, \Omega, \mathcal{O})$, where $\mathcal{A}$ is the action space (hypothesis generation and experiment execution), T is the transition function, R is the reward (scientific value of discovery), and $\mathcal{O}$ is the noisy observation process. Three properties distinguish this from standard control: R cannot be pre-specified (significance is not defined a priori); $\mathcal{S}$ is unbounded; and T is non-stationary (accumulated knowledge changes the exploration frontier). Bayesian Experimental Design [Chaloner and Verdinelli, 1995, Rainforth et al., 2024] provides the normative criterion for the search policy π: select d^* maximizing Expected Information Gain (EIG),

$$d^* = \arg\max_{d \in \mathcal{D}} \ \mathrm{EIG}(d) = \arg\max_{d} \ \mathbb{E}\left[H(\theta) - H(\theta \mid y, d)\right] \tag{1}$$

where $H(\theta)$ is prior entropy over parameters θ and $H(\theta \mid y, d)$ is expected posterior entropy after observing y under d. EIG can be approximated via variational methods [Foster et al., 2019], amortized policies [Foster et al., 2021], and RL [Blau et al., 2022], yet no current AI Scientist system implements it—a gap we identify in Section 8.

3 Related Work

AI Scientist systems. Prior work [Lu et al., 2024, 2026, Schmidgall et al., 2025, Boiko et al., 2023] provides qualitative characterizations of individual systems but no shared framework for cross-system comparison; idea-generation studies [Si et al., 2024, Baek et al., 2024] establish quality baselines without unifying the full discovery loop.

Theoretical foundations. The POMDP framework [Kaelbling et al., 1998] and BED [Chaloner and Verdinelli, 1995, Rainforth et al., 2024] are classical; we are the first to explicitly apply BED as a design target for AI Scientist search policies, connecting POMDP (structural formulation) to BED (normative policy criterion).

Formal models and artifact infrastructure. Taniguchi et al. [2025] propose Collective Predictive Coding as a social model of science; we provide the complementary single-system modular decomposition. Liu et al. [2026] address artifact verification as a V and M_s contribution, discussed in Sections 5.1 and 8.

4 The Servo Framework

We define an AI Scientist system as a tuple (S, G, E, V, M, π) with the following components:

$$\textsc{Servo} = (S,\ G,\ E,\ V,\ M,\ \pi) \tag{2}$$

S**: Search Space** The set of candidate hypotheses, programs, molecules, experimental conditions, or other objects subject to exploration. S may be static or dynamically expanding (open-ended).

$G : M \times S \to h$**: Hypothesis Generator** A function that proposes new candidate $h \in S$ conditioned on accumulated memory M and the current state of the search space. G may involve LLM generation, evolutionary mutation, or symbolic search.

$E : h \times P \to o$**: Experiment Executor** A function that executes hypothesis h under protocol P to produce observation o. The fidelity of E ranges from fast computational simulation (low cost, potential surrogate error) to physical wet-lab execution (high cost, ground truth).

$V : o \to r$**: Validator** A function that assigns a reward signal r to observation o. We distinguish four layers of V:

$$\begin{aligned}
V_{\text{syntax}} &: o \to \{\text{pass}, \text{fail}\} && \text{(deterministic, fast)}\\
V_{\text{semantic}} &: o \to r \in [0, 1] && \text{(LLM-based, stochastic)}\\
V_{\text{empirical}} &: o \to r && \text{(reproduction-based)}\\
V_{\text{human}} &: o \to \{\text{novel}, \text{significant}\} && \text{(non-automatizable)}
\end{aligned}$$

M**: Memory** Structured as (M_e, M_s, M_p). M_e (episodic) stores individual records (h_i, o_i, r_i); M_s (semantic) stores distilled knowledge patterns; M_p (procedural) stores learned protocols. The structure of M directly determines the quality of G and π.

$\pi : b(S) \times M \to h$**: Search Policy** The decision rule for selecting the next hypothesis. The belief state $b(S)$ is updated after each experiment. In the BED interpretation, $\pi^* = \arg\max_d \mathrm{EIG}(d)$ (Equation 1). In practice, current systems use policies ranging from reactive (ReAct) to greedy to tree search.

The operational loop is:

$$\begin{aligned}
&\pi(b, M) \ \to\ G(M_s, S) \ \to\ E(h, P) \ \to\ V(o)\\
&\qquad \to\ M \mathrel{+}= (h, o, r) \ \to\ b(S) \text{ update} \ \to\ \text{repeat}
\end{aligned} \tag{3}$$

We further introduce $H \in [0, 1]$ as an auxiliary parameter denoting the degree of human intervention: $H = 0$ for fully autonomous systems and $H = 1$ for systems in which humans assume the role of G or π.

5 Analysis of Core AI Scientist Systems

We apply SERVO to four end-to-end systems [Boiko et al., 2023, Lu et al., 2024, 2026, Schmidgall et al., 2025] and one artifact infrastructure proposal. Table 1 synthesizes the key dimensions; the cross-system pattern is that every architectural advance is a gain in V completeness, and V completeness is what unlocks closed-loop operation.

	Coscientist	**AI Sci. (2024)**	**AI Sci. (2026)**	**AgentLab**
Novelty measure	None	LLM self-score	3-layer	Human-delegated
Loop closed?	No	Yes (greedy)	Yes (tree)	Partial
π type	ReAct	Greedy	Tree search	Human/seq.
V completeness	V_h only	V_s	$V_e + V_s + V_h$	V_s (biased)
M structure	Ephemeral	M_e	$M_e + M_s$	M_e
H (human role)	G	Low	Low	G+π

Table 1. Comparative SERVO analysis of four core AI Scientist systems. V_s=semantic, V_e=empirical, V_h=human.

5.1 Artifact Infrastructure: Agent-Native Research Artifacts

Liu et al. [2026] address a complementary problem: what the *output* of the AI Scientist loop should look like and how it can be independently verified. Agent-Native Research Artifacts (ARA) replace the narrative paper with a four-layer ontology: `/logic` (claims with falsification criteria), `/src` (executable code), `/trace` (exploration graph: question, decision, experiment, dead_end, pivot), and `/evidence` (raw results). The ARA Seal provides three-level automated verification—V_{syntax} (schema conformance), V_{semantic} (Rigor Auditor, 82.6% detection rate), and $V_{\text{empirical}}$ (sandboxed reproduction)—with V_{human} reserved for significance, novelty, and taste: *"judgments of significance, novelty, and taste are reserved for human reviewers"* [Liu et al., 2026]. The motivation is quantitative: only 45.4% of reproduction requirements are fully specified in published papers, and 90.2% of extension costs are consumed by failed exploration [Liu et al., 2026]. The `/trace` layer addresses M_e and M_s: preserved failure traces accelerated agent performance on extension tasks but also anchored more capable agents to prior solution paths—richer M_e does not monotonically improve π.

6 Domain Analysis

Table 2 maps SERVO instantiations across seven scientific domains; V completeness co-varies with loop closability without exception. To illustrate scale: Aletheia [Feng et al., 2026] attempted 700 Erdős problems, with auditors rating 13/200 graded solutions (6.5%) meaningfully correct; GNoME [Merchant et al., 2023] screened 2.2 million candidate crystal structures via high-throughput DFT. Systems analyzed span mathematics (formal: [Romera-Paredes et al., 2024, Xin et al., 2024]; NL: [Feng et al., 2026]), physics [Udrescu and Tegmark, 2020, Cranmer et al., 2020], chemistry [Bran et al., 2024, Sprueill et al., 2024], materials [Merchant et al., 2023], biology [O'Donoghue et al., 2023], social science [Manning

Domain	V type	V reliability	Loop	π	Primary bottleneck
Math (formal)	V_{formal}	Perfect oracle	Fully closed	RL/MCTS	Pretraining contamination
Math (NL)	$V_{\text{sem}} + V_h$	Imperfect	Partial	Seq. refine	Hallucination; plagiarism
Physics	$V_{\text{emp}} + V_h$	Medium	None/partial	Deterministic	No cross-problem M
Chemistry	V_{emp} (surr.)	Medium	Comp. only	Beam	Wet-lab loop open
Materials	V_{emp} (DFT)	High	Comp. closed	Uncert. samp.	DFT wall-time; synth. lag
Biology	$V_{\text{emp}} + V_h$	Low–med	Partial	None	V metrics insufficient
Social Sci.	Stat. test	Low–med	Partial	Factorial	Circular validity
CS / Algo.	V_{emp}	High	Fully closed	Evol./LLM	Interpretability absent

Table 2. Cross-domain Servo analysis across seven domains. V reliability refers to agreement with ground truth.

et al., 2024, Aher et al., 2023, Park et al., 2023], and CS/algorithms [Real et al., 2020, Sartori and Blum, 2024, Jaber and Jaber, 2026].

The domain pattern is not coincidental. In mathematics (formal), a noise-free V_{formal} unlocks stable RL: the binary, unambiguous reward cannot be hacked. In chemistry and biology, the loop cannot close because no automated V substitutes for physical measurement (NMR, MS, wet-lab validation). Materials science is the single domain where π approximates BED theory: GNoME's uncertainty sampling over a tractable GNN posterior is the closest existing deployment to EIG-optimal search. Social science faces a distinct failure mode: when G and E share the same underlying model, the evaluator systematically confirms the generator's implicit biases [Aher et al., 2023].

7 Illustrative Simulation: Validator Coverage and False Novelty

To make the consequences of V incompleteness concrete, we run a minimal agent-based simulation (20 agents, 200 hypotheses, 40 latent classes, 30 seeds) in which agents estimate novelty via a coverage membership test and share memory only through their communication topology—the primary independent variable. Table 3 formalizes the four conditions as levels of V completeness.

Table 3. Simulation conditions. Each operationalizes a level of V completeness via mean degree k.

Condition	V-completeness	k	Graph	Coverage
Isolated	Incomplete	0	No edges	Own memory only
Lattice	Partial	2	Ring	2-hop neighborhood
Medium	Improved	4	Watts–Strogatz	Local + cross-links
Shared	Complete	$n-1$	Complete	Full community memory

Results. Figure 1 shows FNR (fraction of non-novel hypotheses marked novel) across conditions. Because memory sharing is set-union, broader connectivity structurally reduces FNR—an analytic consequence, not an emergent finding. FNR decreases from 0.955 ± 0.028 (isolated) to 0.008 ± 0.012 (shared); varying classifier accuracy 0.60–0.90 moves FNR by ≤ 0.035, a $27\times$ smaller range than the cross-topology span. A non-trivial consequence emerges at $\geq 10\%$ prior seeding: isolated agents achieve *higher* balanced accuracy than lattice (0.564 vs. 0.529), as connectivity accelerates memory saturation. The analogy to epistemic-network models [Zollman, 2007, 2010] is phenomenological only.

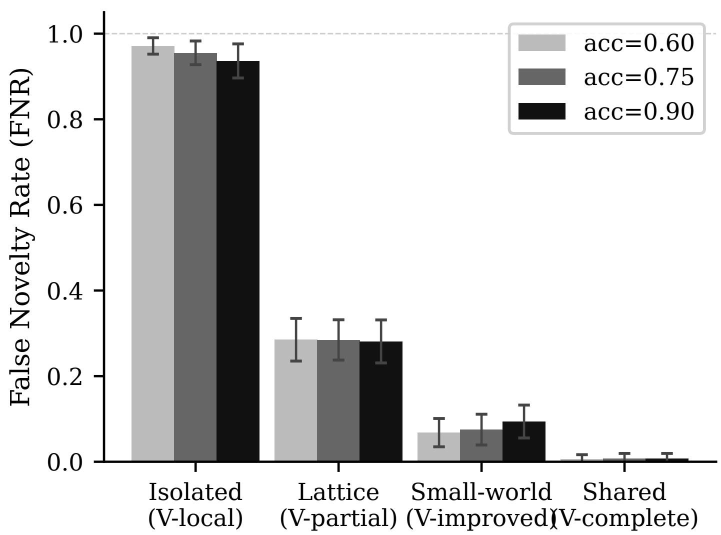

Figure 1. FNR by V-completeness level and classifier accuracy (± 1 std, 30 seeds). Cross-topology span: 0.947; within-condition accuracy range: ≤ 0.035.

Limitations. Synthetic agents with a ground-truth oracle; agents do not generate hypotheses or update beliefs. The simulation is an existence illustration of V incompleteness—the FNR–topology ordering is structurally determined and cannot produce any other outcome.

8 Open Problems

Our analysis identifies three open problems that persist across all systems and domains.

8.1 The Non-Automatability of Novelty

Novelty cannot be reliably evaluated by automated means: every system from Coscientist (no mechanism) to The AI Scientist Nature (three-layer validation) converges on this conclusion. Two failure modes are documented: LLM self-evaluation bias (systematic overestimation; hallucinated results) and pretraining contamination (systems may "discover" results implicit in training data, as Aletheia documents for Erdős problems). The ARA Seal [Liu et al., 2026] demarcates the automated frontier—82.6% detection of structural violations with a 22% blind spot for orphaned experiments—but cannot cross the novelty boundary. Concretely: Agent Laboratory's LLM reviewer over-estimated quality by +2.3 points on average; The AI Scientist (Sakana) hallucinated ablation tables; and Aletheia audited 700 Erdős problems and found only 6.5% of graded solutions meaningfully correct despite surfacing many superficially plausible candidates.

Falsification criterion: Refuted if an automated V achieves human-level novelty discrimination (inter-rater agreement $\geq$0.80) on a contamination-controlled benchmark spanning $\geq$3 distinct scientific domains.

8.2 The BED–Practice Gap for π

Equation 1 specifies the theoretically optimal policy, yet no current system implements it: all deployed policies are heuristic (ReAct, greedy, tree search, human-guided). The partial exception is GNoME [Merchant et al., 2023], whose deep-ensemble uncertainty sampling approximates EIG over a tractable GNN posterior—but only because DFT provides a well-calibrated V and the hypothesis space admits a learnable probabilistic model, conditions absent from LLM-based systems where the "posterior" is implicit in model weights [Foster et al., 2021]. Concretely, every system in Table 1 degrades to heuristic π: Coscientist selects the next tool reactively, The AI Scientist (Sakana) greedily avoids archive similarity, and Agent Laboratory follows a fixed sequential schedule—none improve with additional experimental data, the failure mode EIG-optimal π would avoid.

Falsification criterion: Refuted if any AI Scientist system implements a policy that explicitly computes and maximizes EIG over a posterior $p(\theta \mid y)$ in a domain where the posterior is not tractable from first principles—ruling out tractable-model proxies such as GNoME's deep-ensemble policy.

8.3 The Underdevelopment of Semantic Memory M_s

Current systems store raw experiment records (M_e) but do not distill them into transferable patterns (M_s): a system with 1,000 experiments does not search more intelligently than one with 10. The ARA `/trace` layer [Liu et al., 2026] is the closest approximation, but traces accelerated extension performance while also anchoring capable agents to prior solution paths—M_s structure must be paired with selective forgetting mechanisms to be beneficial. Concretely, GNoME's GNN—one of the few systems where $G(M_s, S)$ genuinely improves—accumulates domain-specific thermodynamic knowledge after 460K DFT calculations, yet this knowledge does not transfer to chemistry or biology; the system is no more capable in any domain it has not explicitly trained on.

Falsification criterion: Refuted if accumulated cross-experiment knowledge demonstrably improves G-generated hypothesis quality across task instances in any single domain, measured by a pre-specified metric.

The three open problems are not independent. A system with complete V and tractable M_s would, by construction, be able to implement an EIG-optimal π: complete V provides a calibrated, hack-resistant reward signal, and developed M_s maintains the hypothesis-space model that EIG requires. They therefore converge on a single structural gap, and progress on any one unlocks disproportionate gains toward the other two. They are also precisely the necessary conditions for a collective AI co-scientist [Taniguchi et al., 2025]:

$V_{\text{collective}}$ requires solved novelty automatability; a principled collective π requires closed BED–practice gap; and the shared semantic memory w requires developed M_s. ARA [Liu et al., 2026] may be the most tractable entry point, simultaneously externalizing M_s and calibrating the sub-novelty layers of V.

9 Discussion

Our analysis establishes a design hierarchy: V choice precedes all other decisions, as it determines closed-loop feasibility and what form π can take. The collective extension [Taniguchi et al., 2025] requires that the three open problems be solved first: $V_{\text{collective}}$ requires novelty automatability; a principled collective π requires closing the BED–practice gap; developed M_s enables the shared semantic memory w. ARA [Liu et al., 2026] may be the most tractable entry point as a structured artifact protocol for w. Limitations: SERVO abstracts domain-specific cost structures of E, non-stationarity of S, and the social layer that Taniguchi et al. [2025] formalize.

10 Conclusion

We introduced SERVO (S, G, E, V, M, π), a formal framework connecting AI Scientist systems to POMDP and BED theory, and applied it to four core systems, one artifact infrastructure proposal, and seven scientific domains. The central finding is that V completeness is the primary determinant of closed-loop feasibility: every architectural advance across surveyed systems is ultimately a gain in V completeness. The three open problems—novelty non-automatability, the BED–practice gap, and M_s underdevelopment—converge on a single organizing question: *how do we build validators and memory structures that are simultaneously complete enough to support closed-loop operation and structured enough to support principled search?*

References

G. V. Aher, R. I. Arriaga, and A. T. Kalai. Using large language models to simulate multiple humans and replicate human subject studies. In *International Conference on Machine Learning*, 2023. arXiv:2208.10264.

J. Baek, S. K. Jauhar, S. Pitis, S. J. Hwang, and J. Teevan. ResearchAgent: Iterative research idea generation over scientific literature with large language models. *arXiv preprint arXiv:2404.07738*, 2024.

T. Blau, E. V. Bonilla, I. Chades, and A. Dezfouli. Optimizing sequential experimental design with deep reinforcement learning. In *International Conference on Machine Learning*, 2022. arXiv:2202.00821.

D. A. Boiko, R. MacKnight, B. Kline, and G. Gomes. Emergent autonomous scientific research capabilities of large language models. *Nature*, 624:570–578, 2023.

A. M. Bran, S. Cox, O. Schilter, C. Baldassari, A. D. White, and P. Schwaller. ChemCrow: Augmenting large-language models with chemistry tools. *Nature Machine Intelligence*, 2024. arXiv:2304.05376.

K. Chaloner and I. Verdinelli. Bayesian experimental design: A review. *Statistical Science*, 10(3):273–304, 1995.

M. Cranmer, A. Sanchez Gonzalez, P. Battaglia, R. Xu, K. Cranmer, D. Spergel, and S. Ho. Discovering symbolic models from deep learning with inductive biases. In *Advances in Neural Information Processing Systems*, 2020. arXiv:2006.11287.

T. Feng, Q. V. Le, T. Luong, T. H. Trinh, G. Bingham, D. Hwang, Y. Chervonyi, et al. Towards autonomous mathematics research. *arXiv preprint arXiv:2602.10177*, 2026. arXiv:2602.10177.

A. Foster, M. Jankowiak, E. Bingham, P. Horsfall, Y. W. Teh, T. Rainforth, and N. Goodman. Variational Bayesian optimal experimental design. In *Advances in Neural Information Processing Systems*, 2019. arXiv:1903.05480.

A. Foster, D. R. Ivanova, I. Malik, and T. Rainforth. Deep adaptive design: Amortizing sequential Bayesian experimental design. In *International Conference on Machine Learning*, 2021. arXiv:2103.02438.

J. Jaber and O. Jaber. AutoKernel: Autonomous GPU kernel optimization via iterative agent-driven search. *arXiv preprint arXiv:2603.21331*, 2026.

J. Jumper, R. Evans, A. Pritzel, T. Green, M. Figurnov, O. Ronneberger, K. Tunyasuvunakool, R. Bates, A. Žídek, A. Potapenko, et al. Highly accurate protein structure prediction with AlphaFold. *Nature*, 596:583–589, 2021.

L. P. Kaelbling, M. L. Littman, and A. R. Cassandra. Planning and acting in partially observable stochastic domains. *Artificial Intelligence*, 101(1–2):99–134, 1998.

R. Liu et al. The last human-written paper: Agent-native research artifacts. *arXiv preprint arXiv:2604.24658*, 2026.

C. Lu, C. Lu, R. T. Lange, J. Foerster, J. Clune, and D. Ha. The AI Scientist: Towards fully automated open-ended scientific discovery. *arXiv preprint arXiv:2408.06292*, 2024.

C. Lu, C. Lu, R. T. Lange, Y. Yamada, S. Hu, J. Foerster, D. Ha, and J. Clune. Towards end-to-end automation of AI research. *Nature*, 651:914–919, 2026. doi: 10.1038/s41586-026-10265-5.

B. S. Manning, K. Zhu, and J. J. Horton. Automated social science: Language models as scientist and subjects. *NBER Working Paper*, (32381), 2024. arXiv:2404.11794.

A. Merchant, S. Batzner, S. S. Schoenholz, M. Aykol, G. Cheon, and E. D. Cubuk. Scaling deep learning for materials discovery. *Nature*, 624:80–85, 2023.

O. O'Donoghue, G. Puthumanaillam, J. Won, P. Moura, M. Oettig, J. M. de Kruif, M. Gini, M. Heinonen, and P. Sloot. BioPlanner: Automatic evaluation of LLMs on protocol planning in biology. *arXiv preprint arXiv:2310.10632*, 2023.

J. S. Park, J. C. O'Brien, C. J. Cai, M. R. Morris, P. Liang, and M. S. Bernstein. Generative agents: Interactive simulacra of human behavior. In *Proceedings of the ACM Symposium on User Interface Software and Technology*, 2023. arXiv:2304.03442.

T. Rainforth, A. Foster, D. R. Ivanova, and F. B. Smith. Modern Bayesian experimental design. *Statistical Science*, 2024. arXiv:2302.14545.

E. Real, C. Liang, D. So, and Q. Le. AutoML-Zero: Evolving machine learning algorithms from scratch. In *International Conference on Machine Learning*, 2020. arXiv:2003.03384.

B. Romera-Paredes, M. Barekatain, A. Novikov, M. Balog, M. P. Kumar, E. Dupont, F. J. R. Ruiz, J. Ellenberg, P. Wang, O. Fawzi, et al. Mathematical discoveries from program search with large language models. *Nature*, 625:468–475, 2024.

C. C. Sartori and C. Blum. Algorithm discovery with large language models: Evolutionary search meets transformer. *arXiv preprint arXiv:2411.00895*, 2024.

S. Schmidgall, Y. Su, Z. Wang, X. Sun, J. Wu, X. Yu, J. Liu, and E. Barsoum. Agent Laboratory: Using LLM agents as research assistants. *arXiv preprint arXiv:2501.04227*, 2025.

C. Si, D. Yang, and T. Hashimoto. Can LLMs generate novel research ideas? A large-scale human study with 100+ NLP researchers. *arXiv preprint arXiv:2409.04109*, 2024.

H. W. Sprueill, C. Edwards, K. Agarwal, M. Olarte, U. Sanyal, C. Johnston, H. Liu, H. Ji, and S. Bhatt. ChemReasoner: Heuristic search over a large language model's knowledge space using quantum-chemical feedback. In *International Conference on Machine Learning*, 2024. arXiv:2402.10980.

T. Taniguchi, S. Takagi, J. Otsuka, Y. Hayashi, and H. T. Hamada. Collective predictive coding as model of science: Formalizing scientific activities towards generative science. *Royal Society Open Science*, 12(6):241678, 2025. doi: 10.1098/rsos.241678. arXiv:2409.00102.

S.-M. Udrescu and M. Tegmark. AI Feynman: A physics-inspired method for symbolic regression. *Science Advances*, 6(16):eaay2631, 2020.

H. Xin, D. Ren, S. Song, Z. Shao, W. Zhao, H. Wang, W. Liu, L. Cheng, et al. DeepSeek-Prover-V1.5: Harnessing proof assistant feedback for reinforcement learning and Monte-Carlo tree search. *arXiv preprint arXiv:2408.08152*, 2024. arXiv:2408.08152.

K. J. S. Zollman. The communication structure of epistemic communities. *Philosophy of Science*, 74(5):574–587, 2007.

K. J. S. Zollman. The epistemic benefit of transient diversity. *Erkenntnis*, 72(1):17–35, 2010.

AI Involvement Checklist

This checklist documents the role of AI systems at each stage of this research, as required by CAISc 2026.

- **Hypothesis development.** The core SERVO framework (S, G, E, V, M, π), the H parameter, the G output spectrum (propositional hypothesis $\rightarrow$ falsifiable construction), and the three open problems were conceived by the human author. Claude Code (Anthropic) elaborated framework components, identified paper-level instantiations of each component, and proposed connections between concepts.
- **Literature search and selection.** The human author determined inclusion criteria and target systems. Claude Code autonomously retrieved and read over 40 full-text PDFs, categorized papers into the taxonomy, and performed claim-level verification against source text.
- **Analysis.** Claude Code extracted S, G, E, V, M, π components for each of the 13 analyzed systems by reading primary sources. The human author reviewed all extractions, corrected mischaracterizations (several substantive corrections were required), and synthesized cross-system patterns.
- **Writing.** Claude Code drafted the majority of this manuscript. The human author provided key conceptual passages (notably §8 and the core definitions in §4), directed structural decisions, and reviewed all drafts.

Overall, the division of labor is best characterized as: the human author provided intellectual direction, framework design, and final judgment; Claude Code provided literature retrieval, analysis execution, and text generation.

Responsibility & Reproducibility Checklist

- **Claims and evidence.** All empirical claims are grounded in cited primary sources. Major claims were verified by direct full-text reading of source PDFs; specific figures (e.g., Science of Science statistics) were cross-checked against institutional press releases where journal access was unavailable.
- **Limitations.** This paper is a conceptual meta-analysis of existing published systems. It contains no original experiments. The SERVO framework is not empirically validated against held-out systems in this submission; its descriptive adequacy rests on consistency across the 13 analyzed systems. Domain-specific cost structures, multi-agent dynamics, and temporal evolution of systems are not modeled.
- **Reproducibility.** The analytical framework can be applied to any AI Scientist system following the component definitions in §4. All analyzed papers are publicly accessible. No proprietary data or closed models were used.
- **Experimental details.** Literature review methodology follows Muka et al. (2020) 24-step systematic review guide. System selection criterion: end-to-end automation of at least one component of the hypothesis–experiment–validation loop.
- **Data and code availability.** Analysis notes will be made available upon acceptance. No datasets were produced.
- **Ethical and societal impact.** The H parameter analysis explicitly addresses the necessity of human oversight in current systems. The open problems discussion (§8) identifies risks of premature convergence under widespread AI Scientist deployment, particularly through the M_s underdevelopment problem.